

A CONGRUENCE BETWEEN MULTINOMIAL COEFFICIENTS

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Let p be a prime number. For every integer $m \neq 0$, let $v(m)$ denote the exponent of the highest power of p which divides m . Put, moreover, $v(0) = \infty$.

For $\mathbf{n} = (n_1, \dots, n_k)$ a k -tuple of integers ≥ 0 , put

$$|\mathbf{n}| = \sum_{i=1}^k n_i, \quad a\mathbf{n} = (an_1, \dots, an_k), \quad X^{\mathbf{n}} = \prod_{i=1}^k X_i^{n_i}, \quad \mathbf{n}! = \prod_{i=1}^k n_i!,$$

and denote by $C(\mathbf{n})$ the coefficient of the monomial $X^{\mathbf{n}}$ in

$$\left(\sum_{i=1}^k X_i \right)^{|\mathbf{n}|}$$

(the multinomial coefficients).

Thus, for every integer $n \geq 0$,

$$\left(\sum_{i=1}^k X_i \right)^n = \sum_{|\mathbf{n}|=n} C(\mathbf{n}) X^{\mathbf{n}}, \quad (1)$$

$$C(\mathbf{n}) = \frac{|\mathbf{n}|!}{\mathbf{n}!}. \quad (2)$$

For $k = 2$ these are the binomial coefficients:

$$C(n_1, n_2) = \binom{n_1 + n_2}{n_1}. \quad (3)$$

Proposition. Let $\mathbf{n}$ be a k -tuple of integers ≥ 0 , let $n = |\mathbf{n}|$, and let $l \geq 1$. If p is a prime number ≥ 5 , then

$$C(p^l \mathbf{n}) \equiv C(\mathbf{n}) \pmod{p^{v(n)+3}}. \quad (4)$$

Example. For $p = 5$, $k = 2$, $l = 1$, and $\mathbf{n} = (1, 1)$, one has

$$\binom{10}{5} = 252 = \binom{2}{1} + 2 \cdot 5^3.$$

Proof. It is known, or follows from (2), that

$$\left(\sum_{i=1}^k X_i \right)^{p^l} = \left(\sum_{i=1}^k X_i^{p^l} \right) + pQ, \quad (5)$$

where Q is a polynomial with integral coefficients, of degree $< p^l$ in each variable. The binomial formula then gives

$$\left(\sum_{i=1}^k X_i \right)^{p^l n} = \left(\sum_{i=1}^k X_i^{p^l} \right)^n + \sum_{i=1}^n p^i \binom{n}{i} R_i, \quad (6)$$

with

$$R_i = \left(\sum_{j=1}^k X_j^{p^l} \right)^{n-i} Q^i. \quad (7)$$

For two polynomials A and B with integral coefficients, we shall write

$$A \doteq B \pmod{r}$$

if the coefficients of $X^{p^l \mathbf{n}}$ in A and B are congruent modulo r . With this notation, (4) is rewritten as

$$\left(\sum_{i=1}^k X_i \right)^{p^l n} \doteq \left(\sum_{i=1}^k X_i^{p^l} \right)^n \pmod{p^{v(n)+3}}. \quad (8)$$

Lemma 1.

$$v \binom{n}{i} \geq v(n) - v(i). \quad (9)$$

Let the additive group $\mathbb{Z}/(n)$ act on itself by translation.¹ If a subset P with i elements of $\mathbb{Z}/(n)$ has as stabilizer² a subgroup H of $\mathbb{Z}/(n)$, $H \cdot P = P$, then P is a union of cosets of H , and $\#H$ divides i . The cardinal

$$r_P = \frac{n}{\#H}$$

of the orbit of P under $\mathbb{Z}/(n)$, in $\mathcal{P}(\mathbb{Z}/(n))$, therefore satisfies

$$p^{v(n)-v(i)} \mid r_P.$$

Hence, since the set of subsets with i elements of $\mathbb{Z}/(n)$ is a disjoint union of orbits,

$$p^{v(n)-v(i)} \mid \binom{n}{i}.$$

Lemma 2. For $p \neq 2, 3$ and $i \geq 3$, one has

$$p^i \binom{n}{i} \equiv 0 \pmod{p^{v(n)+3}}. \quad (10)$$

Indeed,

$$v \left(p^i \binom{n}{i} \right) = i + v \binom{n}{i} \geq v(n) + (i - v(i))$$

by Lemma 1. For $3 \leq i < p$ one has $v(i) = 0$ and $i - v(i) \geq 3$. For $i \geq p$, one has

$$i - v(i) \geq i - \frac{\log i}{\log p}.$$

This last function of i is increasing for $i \geq p$, and is equal to $p - 1 \geq 3$ for $i = p$.

Lemma 3. The coefficient of $X^{p^l \mathbf{n}}$ in R_1 is zero.

None of the monomials $X^{\mathbf{r}}$ appearing in Q has an exponent $\mathbf{r}$ divisible by p^l . The polynomial

$$R_1 = \left(\sum_{j=1}^k X_j^{p^l} \right)^{n-1} Q$$

inherits this property.

It follows from Lemmas 2 and 3 that

$$\left(\sum_{i=1}^k X_i \right)^{p^l n} \doteq \left(\sum_{i=1}^k X_i^{p^l} \right)^n + \binom{n}{2} p^2 R_2 \pmod{p^{v(n)+3}}. \quad (11)$$

¹Thus to every $x \in \mathbb{Z}/(n)$ is associated the permutation of $\mathbb{Z}/(n)$ which sends y to $x + y$.

²If a group G acts on a set E , the stabilizer of a subset P of E is the set of $g \in G$ for which P is stable.

This congruence already implies, by Lemma 1, that

$$C(p^l \mathbf{n}) \equiv C(\mathbf{n}) \pmod{p^{v(n)+2}}. \quad (12)$$

The only monomials $X^{\mathbf{r}}$ with exponent divisible by p^l which occur in $(pQ)^2$ are the monomials

$$X_i^{p^l} X_j^{p^l} \quad (i < j).$$

The coefficient of each of them is

$$\sum_{a \neq 0, p^l} \binom{p^l}{a} \binom{p^l}{p^l - a} = \sum_a \binom{p^l}{a}^2 - 2 = \binom{2p^l}{p^l} - 2, \quad (13)$$

and

$$\binom{2p^l}{p^l} - 2 \equiv \binom{2p}{p} - 2 \pmod{p^3}$$

by (12).

Thus

$$\left(\sum_{i=1}^k X_i \right)^{p^l n} \doteq \left(\sum_{i=1}^k X_i^{p^l} \right)^n + \binom{n}{2} \left[\binom{2p}{p} - 2 \right] R_2 \pmod{p^{v(n)+3}},$$

with R_2 a polynomial with integral coefficients. Since

$$v \binom{n}{2} \geq v(n)$$

by Lemma 1, the proposition follows from the next lemma.

Lemma 4. For p prime ≥ 5 , one has

$$\binom{2p}{p} \equiv 2 \pmod{p^3}.$$

One has

$$p^{-1} \binom{p}{i} \equiv (-1)^{i-1} i^{-1} \pmod{p}$$

for $1 \leq i \leq p-1$, and hence

$$p^{-2} \left[\binom{2p}{p} - 2 \right] = \sum_{i=1}^{p-1} p^{-2} \binom{p}{i}^2 \equiv \sum_{i=1}^{p-1} \frac{1}{i^2} \pmod{p}. \quad (14)$$

Define $z \in \mathbb{Z}/(p)$ by

$$z = \sum_{i \in (\mathbb{Z}/(p))^*} \frac{1}{i^2}.$$

For $u \in (\mathbb{Z}/(p))^*$ one has

$$z = \sum_{i \in (\mathbb{Z}/(p))^*} \frac{1}{(u^{-1}i)^2} = u^2 z$$

and

$$(u^2 - 1)z = 0.$$

Since $p \neq 2, 3$, there exists u such that $u^2 \neq 1$, and $z = 0$. Formula (14) therefore simplifies to

$$p^{-2} \left[\binom{2p}{p} - 2 \right] \equiv 0 \pmod{p},$$

which completes the proof.